Manan Kannasey

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.08
Kendriya Vidyalaya No.1 Indore	Class XII (CBSE)	2021	84.60%
Kendriya Vidyalaya No.1 Indore	Class X (CBSE)	2019	87.20%

SCHOLASTIC ACHIEVEMENTS

Qualified GATE Computer Science 2024 and GATE Data Science & Artificial Intelligence 2025

POSITION OF RESPONSIBILITY

Teaching Assistant -- IIT Hyderabad Aug 2025

- -Introduction to C Programming (Aug-Dec 2025)
- -Database Management Systems (Jan-May 2026)
- -Advanced Data Structures(Aug-Dec 2026)

SKILLS

Languages: Python, C++, C, SQL, LaTeX

ML/Data Science: PyTorch, NumPy, scikit-learn, OpenCV, Pandas, Matplotlib, PyTorch Geometric, Detectron2

Web/Backend: Apache, MySQL

DevOps/Tools: Git, Linux,Docker Compose, VS Code, Wireshark,

RELEVANT COURSES

Visual Computing | Fraud Analytics using Predictive and social Network Techniques | Network Security | Deep Learning | Advanced Computer Network | Machine Learning | Advanced Data Structure

PROJECTS

Multilingual Business Entity Resolution | *Python, scikit-learn, LightGBM, PyTorch, multilingual E5.* GitHub

- Contributed to a pipeline matching **1.73 million business records** against **9.97 million candidate entities** across India, the US, and France, addressing differences in names, scripts, addresses, and record formatting.
- Combined text normalization and TF-IDF candidate retrieval with LightGBM ranking, multilingual E5 cross-encoder scores, and contextual reranking to assess likely matches without comparing every possible record pair.
- Evaluated matching and decision rules against the competition metric, retaining the stronger baseline when later experiments showed no reliable gain; the **team's final submission achieved 0.98076 public macro F0.5**.

Conversational SQL Assistant | *Python, Streamlit, MySQL, Groq API, sqlglot, pytest.* GitHub

- Built a Streamlit-based conversational analytics application that converts natural-language business questions into MySQL queries using a Groq-hosted LLM, database-schema context from INFORMATION_SCHEMA, and recent conversation history.
- Implemented a safety layer with sqlglot AST parsing to permit only one read-only SELECT query; blocked DDL/DML, file output, locking clauses, multi-statement injection, and expensive SQL functions before execution.
- Added numeric LIMIT enforcement with a maximum of 100 returned rows, read-only database access, five-second connection timeouts, safe error handling, and pytest coverage for invalid configuration, malformed model output, unsafe SQL, and query-limit behavior.

Unsupervised Fraud Detection -- BiGAN vs. AnoGAN | *Python, PyTorch, scikit-learn, NumPy, Pandas* GitHub

- Built BiGAN and AnoGAN anomaly-detection pipelines for highly imbalanced credit-card transactions, training on legitimate transactions only to identify fraud as deviation from normal transaction patterns.
- Implemented robust feature preprocessing, normal-only training splits, adversarial training with gradient penalty, and composite anomaly scores combining reconstruction, latent-space, and critic-based signals.
- Calibrated detection thresholds on validation data and evaluated the final models on a held-out test set; AnoGAN achieved **F1-score 0.796** and **ROC-AUC 0.968**, while BiGAN provided a higher-precision alternative for lower false-alert volume.

EXTRACURRICULAR

FOOTBALL | CHESS| SWIMMING

MASTER'S THESIS

Robustness of GNN-Based Network Intrusion Detection Systems | Under Prof. Saurabh Kumar.

- Investigated the robustness of **GNN-based Network Intrusion Detection Systems (NIDS)** against problem-space **structural adversarial attacks** through literature review and experimental study.
- Developed an **E-GraphSAGE**-inspired flow-graph research pipeline representing **IP:port** endpoints as nodes and NetFlow records as attributed edges for intrusion detection.
- Examined C2xB, C2xM, and add-node attack settings, and studied adversarial training as a baseline approach for improving structural robustness.
- Formulated **GUARD-E**, an **uncertainty- and neighbourhood-consistency-gated message**-passing approach; designed evaluation using **F1-score**, false-positive rate, attack-detection rate, and attack-budget analysis.